• Viviparous mode of seed germination is found in halophytes

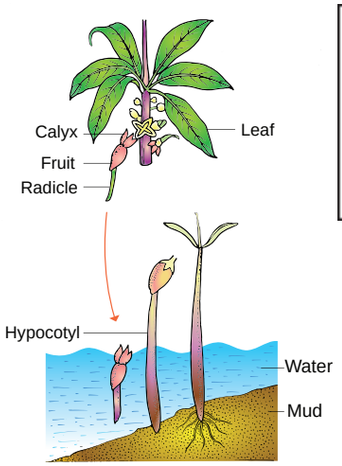


Figure 6.27: Viviparous type of seed germination

Anatomical adaptations

- Epidermal cells of stem is heavy cutinized, almost squarish and are filled with oil and tannins.
- **'Star' shaped** sclereids and **'H' shaped** heavy thickened spicules that provide mechanical strength to cortex are present in the stem.
- The leaves may be dorsiventral or isobilateral with **salt secreting glands**.

Physiological adaptations

- High osmotic pressure exists in some plants .
- Seeds germinate in the fruits while on the mother plant (**Vivipary).**

DO YOU KNOW?

Out of three districts of Tamil Nadu (Nagapattinam, Thanjavur and Thiruvarur), Muthupet (Thiruvarur district) was less damaged by Gaja cyclone (November 2018) due to the presence of mangrove forest.

6.4 Dispersal of Fruits and Seeds

Both fruits and seeds possess attractive colour, odour, shape and taste needed for the dispersal by birds, mammals, reptiles, fish, ants and insects even earthworms. The seed consists of an embryo, stored food material and a protective covering called **seed coat**. As seeds contain miniature but dormant future plants, their dispersal is an important criterion for distribution and establishment of plants over a wide geographical area. The dissemination of seeds and fruits to various distances from the parent plant is called seed and fruit dispersal. It takes place with the help of ecological factors such as wind, water and animals.

Seed dispersal is a regeneration process of plant populations and a common means of colonizing new areas to avoid seedling level competition and from natural enemies like herbivores, frugivores and pathogens.

Fruit maturation and seed dispersal is influenced by many ecologically favourable conditions such as Season (Example: Summer), suitable environment, and seasonal availability of dispersal agents like birds, insects etc.

Seeds require agents for dispersal which are crucial in plant community dynamics in many ecosystems around the globe. They offer many benefits to communities such as food and nutrients, migration of seeds across habitats and helps spreading plant genetic diversity.

6.4.1 Dispersal by Wind (Anemochory)

The individual seeds or the whole fruit may be modified to help for the dispersal by wind. Wind dispersal of fruits and seeds is quite common in tall trees. The adaptation of the wind dispersed plants are

- **Minute seeds:** Seeds are minute, very small, light and with inflated covering. Example: Orchids.
- **Wings:** Seeds or whole fruits are flattened to form a wing. Examples: Maple, *Gyrocarpus, Dipterocarpus* and *Terminalia*



Figure 6.28: *Asclepias*



Figure 6.29: *Gyrocarpus*

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